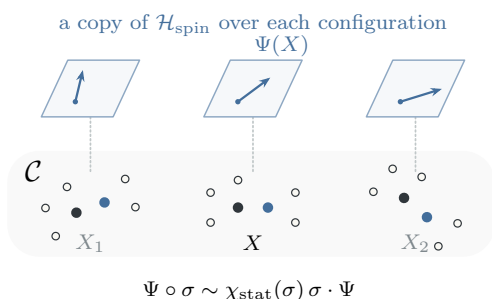


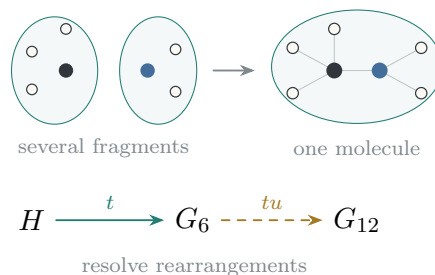
Quantum State Spaces of Nonrigid Molecules and their Complexes

Isaac AshLind

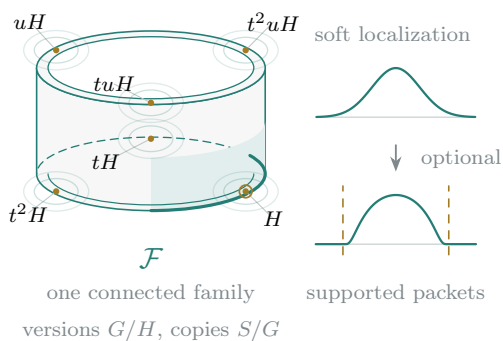
§§ 1–3 Nuclei and spin



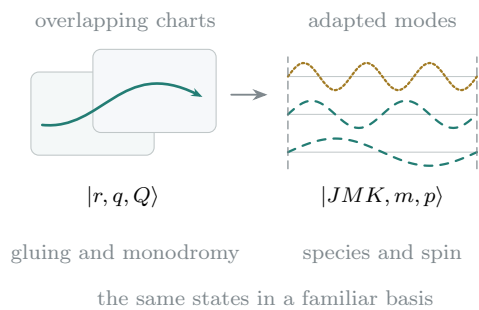
§§ 4–6 Groupings and symmetry



§§ 7–9 Localized states and geometry



§§ 10–12 Position and momentum



Geometric abstract. Each configuration X carries a spin fiber $\mathcal{H}_{\text{spin}}$, with value $\Psi(X)$. Choose groupings and feasible motions. Build localized states and feasible families, then describe the same states in position and momentum. Hard support is optional. Chart gluing preserves the distinction between separate feasible components.

1 Configuration Space

Consider N labeled nuclei in Euclidean space. Take the *configuration space* to be the set of all linear maps $X : \mathbb{R}^N \rightarrow \mathbb{R}^3$ that place the center of mass at the origin. In a Cartesian basis, represent X by a $3 \times N$ matrix whose i th column is the position of nucleus i

$$X = \begin{pmatrix} x_1 & x_2 & \cdots & x_N \\ y_1 & y_2 & \cdots & y_N \\ z_1 & z_2 & \cdots & z_N \end{pmatrix}.$$

The column vector $\mathbf{m} = (m_1, \dots, m_N)^\top$ contains the nuclear masses. The centered configuration space is

$$\mathcal{C} = \{X \in \text{hom}(\mathbb{R}^N, \mathbb{R}^3) : X\mathbf{m} = 0\}, \quad \dim \mathcal{C} = 3N - 3.$$

The nuclei are sorted into types by their charge, mass, and spin, with like nuclei permutable. We also allow global inversion. For types indexed by α , each having multiplicity n_α , the overall permutation and inversion group¹ is taken as

$$S = \left(\prod_{\alpha} S_{n_\alpha} \right) \times \{E, E^*\}.$$

The inversion group $\{E, E^*\}$ is isomorphic to C_2 . We adopt the molecular literature's convention where a star indicates inversion. Each element $s \in S$ is therefore a permutation σ or a permutation–inversion σ^* .

As matrices, the $X \in \mathcal{C}$ form a vector space on which permutations act by multiplication from the right

$$s \cdot X = XP_s, \quad P_{\sigma^*} = -P_\sigma.$$

Here P_σ reorders the columns according to

$$\sigma \cdot (\mathbf{x}_1, \dots, \mathbf{x}_N) = (\mathbf{x}_{\sigma^{-1}(1)}, \dots, \mathbf{x}_{\sigma^{-1}(N)}).$$

With this convention,

$$s \cdot (t \cdot X) = (XP_t)P_s = XP_tP_s = (st) \cdot X.$$

The matrices satisfy $P_{st} = P_tP_s$. Multiplication by these matrices on the right therefore defines a left action of S on $\mathcal{C}$.

We can also act on X from the left with rotations $r \in SO(3)$ via 3×3 rotation matrices

$$r \cdot X = RX.$$

The actions commute by associativity

$$r \cdot (s \cdot X) = RXP_s = s \cdot (r \cdot X).$$

We drop the dots when the action is understood.

¹The complete nuclear permutation–inversion (CNPI) group [1]; see also Longuet-Higgins [2].

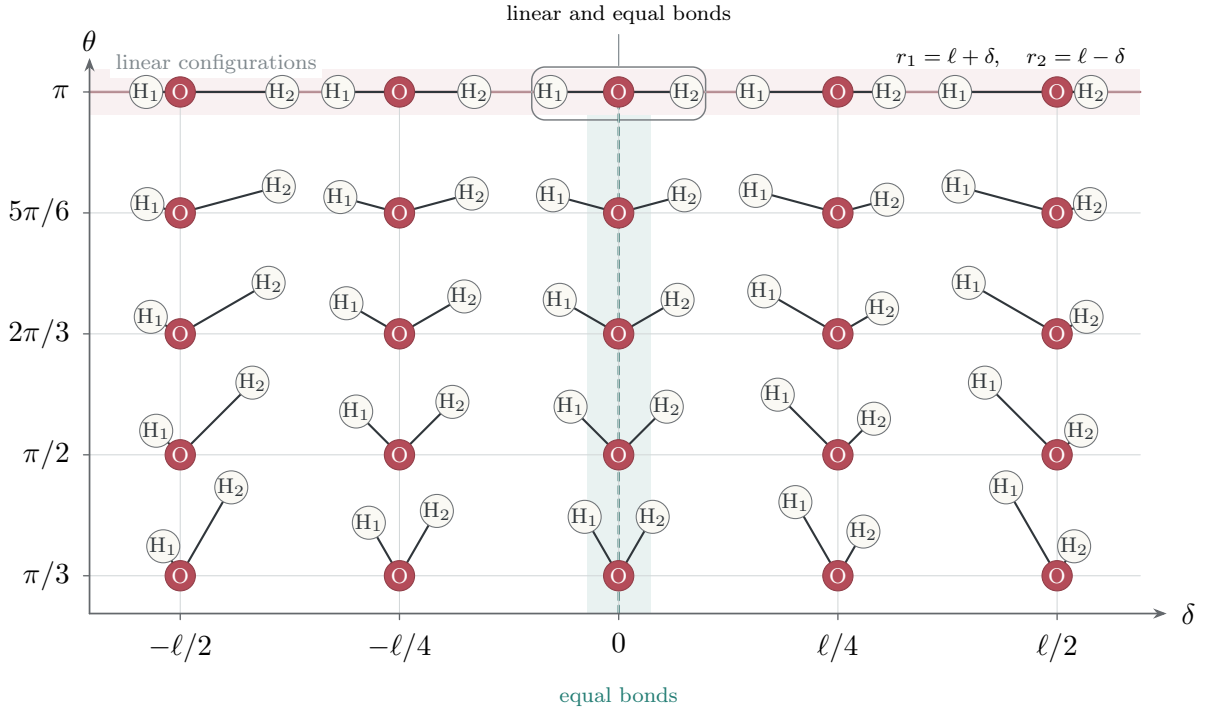


Figure 1: A two-coordinate water family in $\mathcal{C}$, with each configuration centered at its mass center. Fix $r_1 + r_2 = 2\ell$ and vary $\delta = (r_1 - r_2)/2$ and the bond angle θ . At $\delta = 0$, exchanging the hydrogens is equivalent to a rigid rotation. The linear boundary $\theta = \pi$ admits rotations about the molecular axis that fix every nucleus. Both symmetries occur at their intersection. See Section 5 for orbit stabilizers and Section 10 for the axial redundancy of linear configurations. Overall orientation is suppressed.

2 Ambient Hilbert Space

The *ambient Hilbert space* for our configuration space, including the spin of the nuclei, is

$$\mathcal{H}_{\text{ambient}} = L^2(\mathcal{C}) \otimes \mathcal{H}_{\text{spin}}, \quad \mathcal{H}_{\text{spin}} = \bigotimes_{i=1}^N \mathbb{C}^{2I_i+1}.$$

Here $L^2(\mathcal{C})$ uses the translation-invariant volume measure dX up to normalization. The i th nucleus has spin number I_i , and its spin vectors lie in $\mathbb{C}^{2I_i+1}$. A general vector in $\mathcal{H}_{\text{spin}}$ is a linear combination of products $\xi_1 \otimes \cdots \otimes \xi_N$, with factors ordered by nuclear label.

The group S permutes like-nucleus spin factors in the same order as the configuration columns, with inversion acting trivially

$$s \cdot (\xi_1 \otimes \cdots \otimes \xi_N) = \xi_{\sigma^{-1}(1)} \otimes \cdots \otimes \xi_{\sigma^{-1}(N)},$$

where σ is the permutation underlying s . The action extends linearly to all spin vectors.

A wavefunction is a square-integrable map

$$\Psi : \mathcal{C} \rightarrow \mathcal{H}_{\text{spin}}, \quad \|\Psi\|^2 = \int_{\mathcal{C}} \|\Psi(X)\|_{\mathcal{H}_{\text{spin}}}^2 dX < \infty.$$

For example, if $\xi, \eta \in \mathcal{H}_{\text{spin}}$ are orthonormal and $\Psi(X) = a(X)\xi + b(X)\eta$, then

$$\|\Psi(X)\|_{\mathcal{H}_{\text{spin}}}^2 = |a(X)|^2 + |b(X)|^2, \quad \|\Psi\|^2 = \int_{\mathcal{C}} (|a(X)|^2 + |b(X)|^2) dX.$$

As in $L^2(\mathbb{R}^n)$, we identify wavefunctions whose difference has zero norm

$$\Psi \sim \Phi \iff \int_{\mathcal{C}} \|\Psi(X) - \Phi(X)\|_{\mathcal{H}_{\text{spin}}}^2 dX = 0.$$

3 Physical Hilbert Space

Exchange statistics select the *physical Hilbert space* $\mathcal{H} \subseteq \mathcal{H}_{\text{ambient}}$.

Inversion imposes no additional exchange-statistics constraint, so we pass to $S/\langle E^* \rangle$, identifying each coset $\{\sigma, \sigma^*\}$ with its underlying permutation σ . Let σ_α be its restriction to nuclei of type α , with spin number I_α . The statistics character is

$$\chi_{\text{stat}}(\sigma) = \prod_{\alpha} (\text{sgn } \sigma_\alpha)^{2I_\alpha}.$$

Recall that $\Psi(X)$ is a spin vector and σ permutes spin factors. Write $(\Psi \circ \sigma)(X) = \Psi(\sigma X)$ and $(\sigma \cdot \Psi)(X) = \sigma \cdot \Psi(X)$. Using the equivalence relation defined above, the physical Hilbert space is

$$\mathcal{H} = \{\Psi \in \mathcal{H}_{\text{ambient}} : \Psi \circ \sigma \sim \chi_{\text{stat}}(\sigma) \sigma \cdot \Psi, \quad \forall \sigma \in S/\langle E^* \rangle\}.$$

In words, the physical wavefunctions are those which respect exchange statistics under permutations in the configuration space, with the spin factors reorganized accordingly.

The parity sectors of the physical Hilbert space are

$$\mathcal{H}^\pm = \{\Psi \in \mathcal{H} : \Psi \circ E^* \sim \pm \Psi\},$$

where $(\Psi \circ E^*)(X) = \Psi(-X)$. They give the orthogonal decomposition

$$\mathcal{H} = \mathcal{H}^+ \oplus \mathcal{H}^-, \quad \Psi = \Psi_+ + \Psi_-, \quad \Psi_\pm(X) = \frac{1}{2}(\Psi(X) \pm \Psi(-X)).$$

Extend χ_{stat} to S by ignoring the star, and define the parity characters by

$$\chi_\pm(\sigma) = 1, \quad \chi_\pm(\sigma^*) = \pm 1.$$

A physical wavefunction of parity $\pm$ then obeys

$$\Psi_\pm \circ s \sim \chi_{\text{stat}}(s) \chi_\pm(s) s \cdot \Psi_\pm, \quad \forall s \in S.$$

Here composition acts on the configuration argument, while $s \cdot \Psi_\pm$ acts on the spin factors.

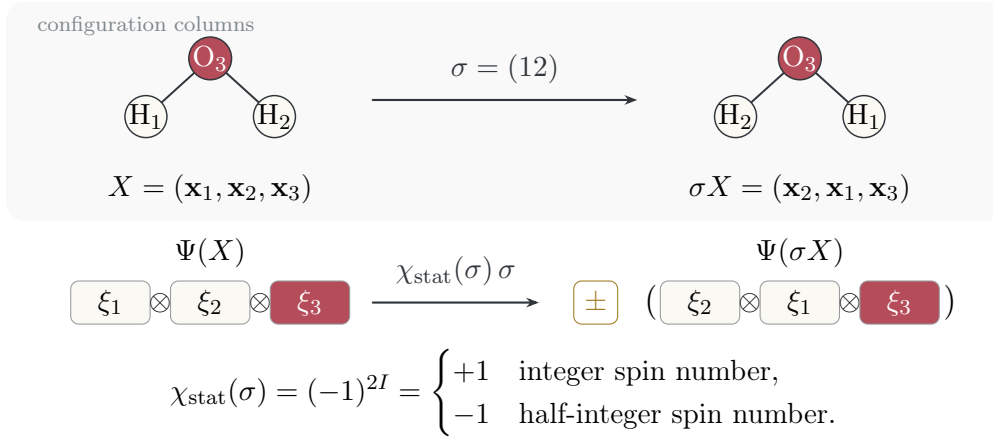


Figure 2: For $\sigma = (12)$, the physical-state condition relates the spin vectors at X and σX by exchanging factors and multiplying the whole vector by $\chi_{\text{stat}}(\sigma)$. Here I is the common spin number of nuclei 1 and 2. A product value is shown, and the rule extends linearly to all spin vectors. The relation holds almost everywhere, as expressed by $\Psi \circ \sigma \sim \chi_{\text{stat}}(\sigma) \sigma \cdot \Psi$.

4 Molecules and Complexes

- Partition the nuclei into fragments. One molecule is the one-fragment case. Relative fragment positions and orientations remain in $\mathcal{C}$.
- For $\text{KRb} + \text{KRb}$, label potassium 1, 2 and rubidium 3, 4.

$$\text{KRb} + \text{KRb} \longrightarrow \text{K}_2\text{Rb}_2 \longrightarrow \text{K}_2 + \text{Rb}_2, \quad S = \langle (12), (34), E^* \rangle \cong C_2^3.$$

$$G_{\text{in}} = \langle (12)(34), E^* \rangle, \quad G_{\text{K}} = \langle (12), E^* \rangle, \quad G_{\text{Rb}} = \langle (34), E^* \rangle.$$

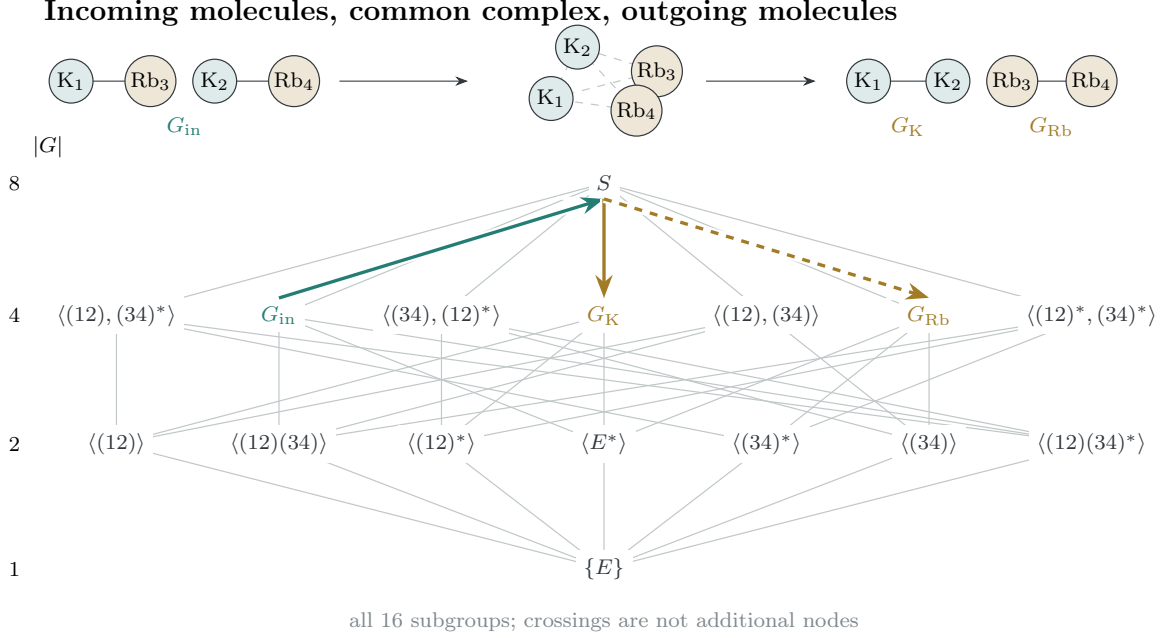


Figure 3: All 16 subgroups of $S = \langle (12), (34), E^* \rangle$. The highlighted routes ascend from the incoming group to S and descend to the individual product groups. Heights give $|G|$. Version counts also require a reference geometry. The routes show group inclusions, with accessibility to be supplied separately.

- Each channel group has order 4. Any two generate S . Separate group inclusion from physical accessibility.
- For $^{40}\text{K}^{87}\text{Rb}$, experiment finds near-conservation of nuclear spin and corresponding product rotational parities.² Symmetry gives allowed sectors, not yields.
- Choose a reference geometry to count versions.

5 Symmetry Groups

The construction so far accommodates all possible arrangements of nuclei without imposing energetic or experimental restrictions. Experimental resolution informs the choice of a feasible regime with a group $G \leq S$ of feasible operations. For a given configuration X , there is a natural minimal symmetry H consisting of the permutation-inversions equivalent to rotations in $SO(3)$. By looking at the subgroup lattice for $H \leq G \leq S$ along with a reference configuration and chosen paths, we can carve out the shapes of feasible regimes and their symmetries without first specifying a potential. This is a kinematic model. Which regimes are accessible in practice is left to experiment.

Choose a reference configuration X_0 and look at its orbit under rigid rotations,

$$SO(3)X_0 = \{rX_0 : r \in SO(3)\}.$$

²Hu et al. [3].

Its point group is the subgroup

$$H = \{h \in S : hX_0 \in SO(3)X_0\}.$$

The point group H stabilizes the rotational orbit of X_0 . Thus, for every $h \in H$, there is a rotation $R_h \in SO(3)$ such that

$$hX_0 = R_hX_0.$$

For nonlinear X_0 , this rotation is unique. The rotational orbit $SO(3)X_0$ describes the rigid reference, and $G = H$ is the minimal feasible regime.

A rearrangement is described by a continuous path

$$\gamma : [0, 1] \rightarrow \mathcal{C}$$

beginning at X_0 and ending at a rotated, permutation-inverted copy of X_0 ,

$$\gamma(0) = X_0, \quad \gamma(1) = rsX_0,$$

for some $r \in SO(3)$ and $s \in S$. The motion is nonrigid if the path leaves $SO(3)X_0$.

If the resolved motions correspond to $s_1, \dots, s_k \in S$, they generate the feasible group

$$G = \langle H, s_1, \dots, s_k \rangle, \quad H \leq G \leq S.$$

The cosets G/H are the versions³ accessible within the regime.

Adding an operation s moves from G to $\langle G, s \rangle$.

6 Methylamine

- Take CH_3NH_2 with five hydrogens of the same isotope. Label methyl hydrogens 1, 2, 3, amino hydrogens 4, 5, carbon 6, and nitrogen 7.
- Choose the reference mirror plane through 1, 6, 7, exchanging 2 with 3 and 4 with 5.

$$S = S_5 \times \{E, E^*\}, \quad |S| = 240, \quad H = \langle b \rangle, \quad b = (23)(45)^*.$$

- Keeping bonds restricts the candidates to $[H, B]$. Here S_A permutes the labels in A .

$$B = (S_{\{1,2,3\}} \times S_{\{4,5\}}) \times \{E, E^*\}, \quad |B| = 24, \quad t = (123), \quad u = (45).$$

$$G_6 = \langle H, t \rangle, \quad |G_6/H| = 3, \quad G_{12} = \langle H, t, u \rangle, \quad |G_{12}/H| = 6.$$

- Above G_6 , adding u , E^* , or u^* gives three order-12 candidates. Any two extensions generate B .
- Microwave and far-infrared spectra resolve torsion and inversion in the G_{12} regime.⁴ The amino inversion includes a corrective 60° methyl turn, represented by tu .⁵
- Compare the candidates through species and spin weights. The full interval $[H, S]$ has 36 subgroups, compared with 10 in $[H, B]$.

³The term *version* follows Bunker and Jensen [4]; the full set of versions is S/H , while G/H gives those accessible within the regime.

⁴Ilyushin et al. [5] and Kręglewski [6].

⁵Kleiner and Hougen [7].

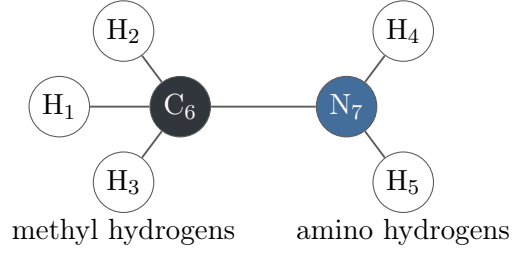


Figure 4: Nuclear labels and bonds for methylamine. All five hydrogens are like nuclei in S . The drawing specifies connectivity, with the three-dimensional geometry suppressed.

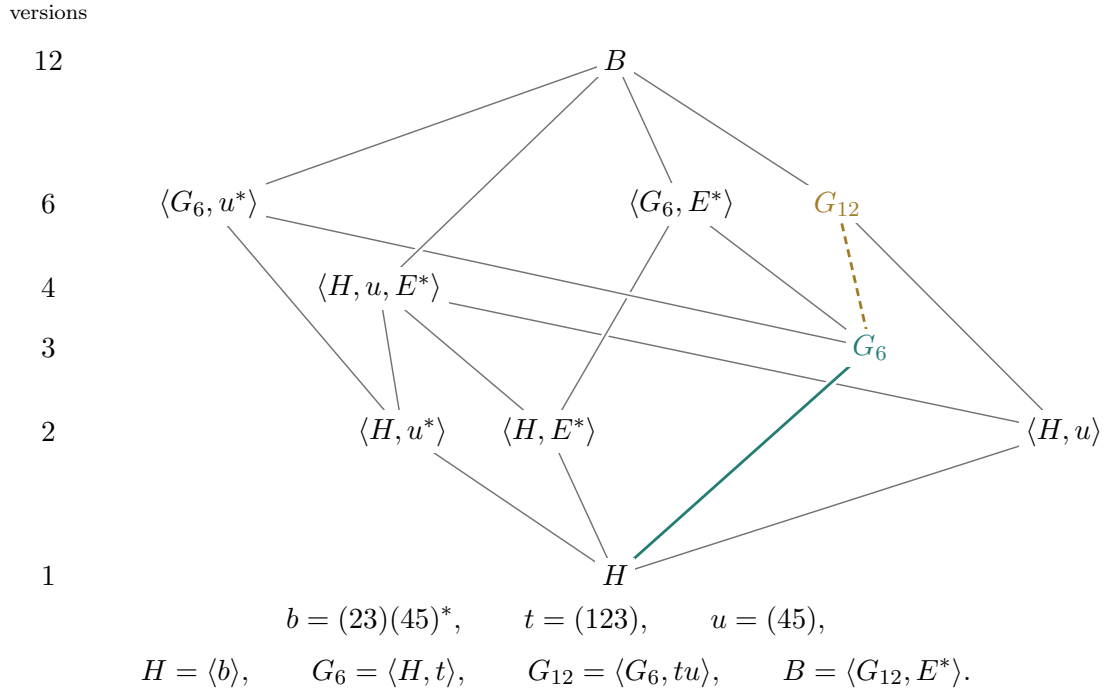


Figure 5: The bond-preserving subgroup interval for methylamine. Numbers at left give $|G/H|$ at each height. The solid teal edge adds methyl torsion and the dark dashed edge adds the coupled torsion–inversion extension. Crossings are not additional nodes.

7 Versions and Symmetry Species

- The irreducible representations of G are its *species*. Pair spatial species with nuclear spin and compute spin weights.
- Choose states at one version, specify their H action, and induce over G/H . Start with a local vibrational ground state at fixed rotational angular momentum.
- Follow species across subgroup inclusions. For complexes, induce incoming channel states and restrict to product groups.
- Use double cosets to organize connections between versions.
- Project a spatial state onto each species. Write $\chi_\Gamma = \text{tr } \Gamma$ and $d_\Gamma = \dim \Gamma$. The projector retains all copies of that species.

$$(U_g f)(X) = f(g^{-1}X), \quad P_\Gamma = \frac{d_\Gamma}{|G|} \sum_{g \in G} \overline{\chi_\Gamma(g)} U_g.$$

$$w_\Gamma(f) = \frac{\|P_\Gamma f\|^2}{\|f\|^2}, \quad \sum_\Gamma w_\Gamma(f) = 1, \quad f \neq 0.$$

- One scalar state fixed by H gives methylamine's three torsional version states.

$$\mathbb{C}[G_6/H] \cong A_1 \oplus E.$$

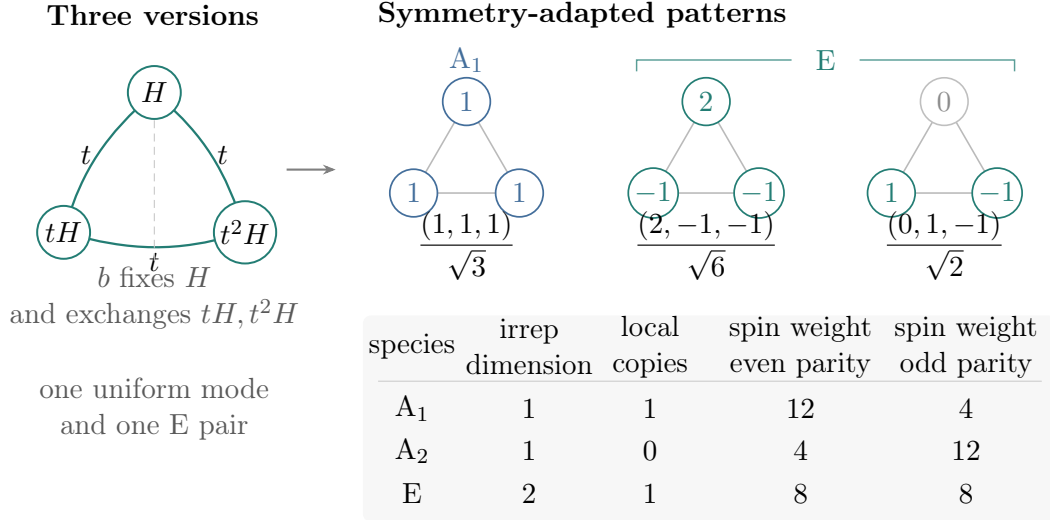


Figure 6: Three scalar version states give one A_1 mode and one E pair. Spin weights count the five proton spins at fixed total inversion parity.

8 Localized States

- In G_6 , choose a normalized real scalar packet g_0 , fixed by H and localized near the reference rotational orbit. Its translates sit at H, tH, t^2H .

$$g_j(X) = g_0(t^{-j}X), \quad j = 0, 1, 2.$$

- All pairwise overlaps equal $c = \langle g_0, g_1 \rangle$. The squared spatial projection shares of g_0 are

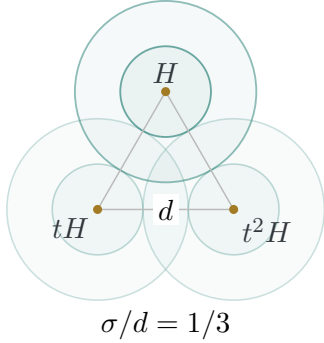
$$w_{A_1} = \frac{1+2c}{3}, \quad w_E = \frac{2(1-c)}{3}, \quad w_{A_2} = 0.$$

- Independent packets span $A_1 \oplus E$. For a planar illustration, take equal Gaussians of position variance σ^2 on an equilateral triangle of side d .

$$c = \exp\left(-\frac{d^2}{8\sigma^2}\right).$$

- Add u -related packets for the six versions in G_{12}/H . Resolve the u action through sums and differences.
- Pair with spin and impose the feasible support restriction of Section 9, when required.
- Treat symmetry-enforced nodes with the required pointwise regularity.

(a) Localized versions



(b) Spatial species shares

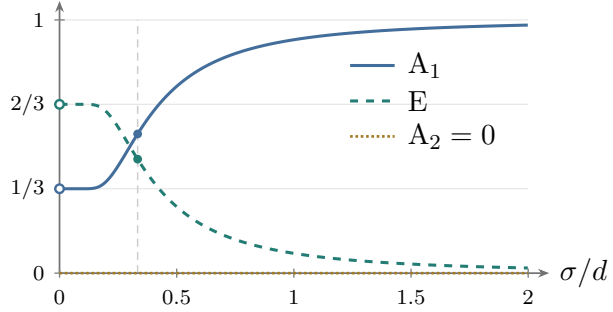


Figure 7: A planar Gaussian model of methylamine’s three torsional version packets. The action of t, b is the same as in Fig. 6. Contours have radii σ and 2σ . The guide marks the width at left. Narrow packets approach shares $(1/3, 2/3, 0)$ in (A_1, E, A_2) , and broad packets approach $(1, 0, 0)$. Each width specifies a different packet.

9 Feasible Configuration Space

- Sweep chosen rearrangement paths through rotations and G . Thicken in the normal directions to obtain a connected region $\mathcal{F} \subseteq \mathcal{C}$.

- Require $s\mathcal{F} = \mathcal{F}$ for $s \in G$ and $s\mathcal{F} \cap \mathcal{F} = \emptyset$ otherwise. The copies are indexed by S/G , with $|G/H|$ reference versions in each.

$$S/H \longrightarrow S/G, \quad sH \longmapsto sG.$$

- For methylamine's G_{12} regime, sweep the torsional circle along the umbrella coordinate. Picture a short tube $S^1 \times [-1, 1]$, thickened by normal displacements, with orientations over each shape.
- Check the product assumption, connectedness, and separation. Recover the rigid limit.

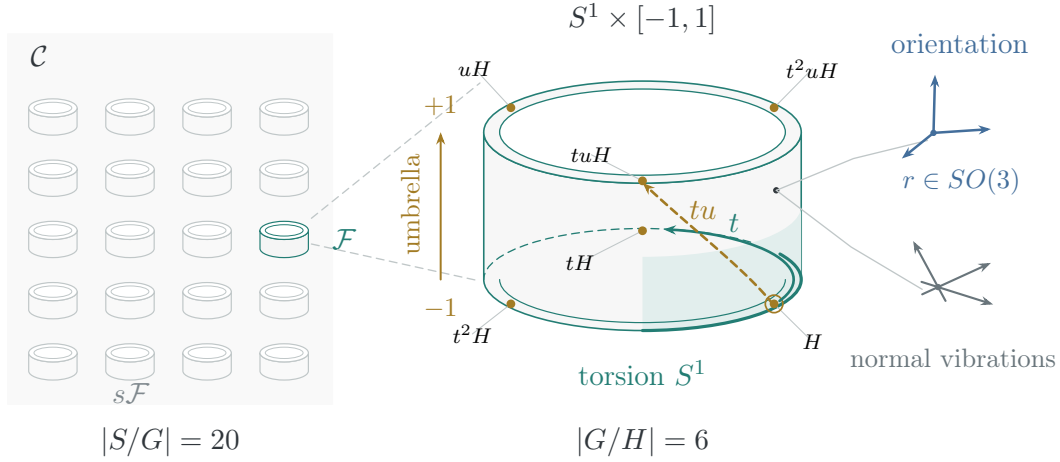


Figure 8: A torsion–umbrella family for G_{12} . Torsion runs around the tube and the umbrella coordinate along it. Thickness represents normal displacements. Each marked version carries its orientations. The highlighted H cell is followed into the position chart of Fig. 9 and the six-cell layout of Fig. 10. The twenty copies in $\mathcal{C}$ each contain six reference versions.

- For an exact support restriction, use the whole symmetry-related union.

$$\mathcal{H}_{\mathcal{F}} = \left\{ \Psi \in \mathcal{H} : \Psi = 0 \text{ a.e. outside } \bigcup_{s \in S} s\mathcal{F} \right\}.$$

- For open regions, compactly supported smooth packets, combined with spin and projected onto exchange statistics, span this space densely. Gaussians retain tails.
- Nested regions give nested state spaces. Group inclusion alone does not specify geometric inclusion.

10 Position Representation

- Choose a body frame and a mass-orthogonal normal frame $e_a(q)$. Use r for orientation, q for internal shape, and Q for normal coordinates.

$$\varphi(r, q, Q) = r \left(X_0(q) + \sum_{a=1}^d Q_a e_a(q) \right), \quad \psi(r, q, Q) = \Psi(\varphi(r, q, Q)).$$

- On a regular chart, pull back the configuration measure. Here dr is normalized Haar measure and $\mathcal{J}$ is the chart density.

$$d\mu = \mathcal{J}(q, Q) dr dq dQ, \quad \|\Psi\|_{\varphi(\mathcal{U})}^2 = \int_{\mathcal{U}} \|\psi(r, q, Q)\|_{\mathcal{H}_{\text{spin}}}^2 d\mu.$$

- The generalized states $|r, q, Q\rangle$ resolve the chart identity with $d\mu$.
- Write $x = (r, q, Q)$. Carry each configuration action through the source and target charts. For $s \in S$, solve for $x' = T_s x$.

$$T_s = \varphi_{\text{target}}^{-1} \circ s \circ \varphi_{\text{source}},$$

- On a symmetry-invariant reference family, first find the new shape q' and compensating rotation $A_s(q)$. Expand the transformed normal frame in the target frame to obtain $M_s(q)$.

$$\begin{aligned} X_0(q)P_s &= A_s(q)X_0(q'), \\ e_a(q)P_s &= A_s(q) \sum_b e_b(q') [M_s(q)]_{ba}, \end{aligned}$$

$$T_s(r, q, Q) = (rA_s(q), q', M_s(q)Q).$$

- This is exact on the normal chart. Finding q' uses the chosen shape coordinates, and the normal-frame expansion is linear algebra.
- A lab rotation R gives $T_R(r, q, Q) = (Rr, q, Q)$. The induced spatial operator is the inverse pullback. Permuting the spin factors gives the physical-state condition.

$$\begin{aligned} (U_s \psi)(x) &= \psi(T_s^{-1}x), \quad U_s |x\rangle = |T_s x\rangle. \\ \psi(T_\sigma x) &\sim \chi_{\text{stat}}(\sigma) \sigma \cdot \psi(x). \end{aligned}$$

- With $\tilde{\psi} = \mathcal{J}^{1/2}\psi$, the same operator in the measure $dr dq dQ$ includes the density ratio.

$$(\tilde{U}_s \tilde{\psi})(x) = \left(\frac{\mathcal{J}(x)}{\mathcal{J}(T_s^{-1}x)} \right)^{1/2} \tilde{\psi}(T_s^{-1}x).$$

- Use these formulas wherever the source and target charts apply. Treat axial redundancy at linear configurations.
- For an explicit methylamine model, take the C–N axis as body z and use η for the umbrella coordinate. In a periodic body frame, tu gives a half-turn of the frame, a $-\pi/3$ torsional shift, and $\eta \mapsto -\eta$.⁶

⁶One reference family places the amino hydrogens at $(\eta, \pm a, z_A)$ and the methyl hydrogens at angles $\tau - 2\pi(k-1)/3$, $k = 1, 2, 3$, at fixed radius and height about the C–N axis. Center the whole configuration as before.

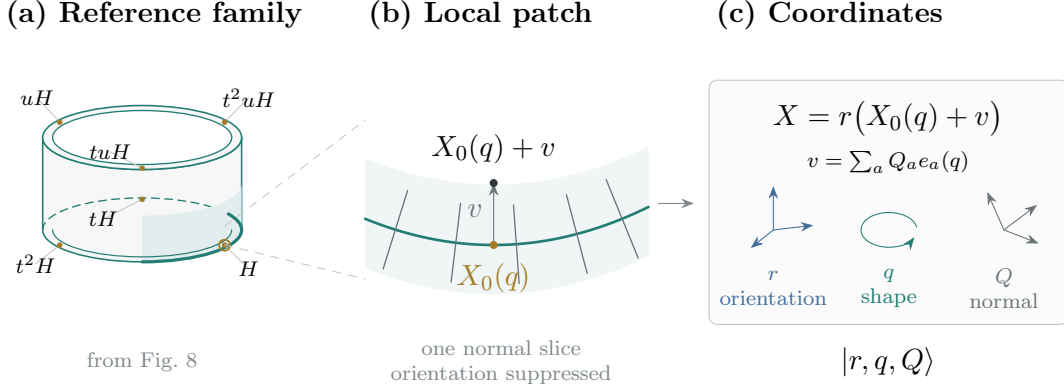


Figure 9: A local patch of the same highlighted cell. The middle panel enlarges a normal slice through the family in Fig. 8. The displacement v is mass-orthogonal to the rotation-swept family. The coordinates restore the suppressed orientation and normal directions, ready for the gluing in Fig. 10.

- Twist the body frame by $R_z(-\rho\tau)$, with ρ a chosen frame parameter. Retain two normal coordinates Q_+, Q_- chosen to transform evenly and oddly under tu . These are symmetry-adapted coordinates, not vibrational eigenmodes.

$$\alpha = \pi - \frac{\rho\pi}{3}, \quad x = (r, \tau, \eta, Q_+, Q_-),$$

$$T_{tu}x = (rR_z(\alpha), \tau - \pi/3, -\eta, Q_+, -Q_-),$$

$$(U_{tu}\psi)(x) = \psi(rR_z(-\alpha), \tau + \pi/3, -\eta, Q_+, -Q_-).$$

- Six applications close through the seam of Section 12. For proton exchange, $\chi_{\text{stat}}(tu) = -1$.

$$T_{tu}^6x = (rR_z(-2\pi\rho), \tau - 2\pi, \eta, Q_+, Q_-) \sim x, \quad \psi(T_{tu}x) \sim -(tu) \cdot \psi(x).$$

- General normal coordinates replace the signs by matrices. The rotational factor records the frame choice. It does not prescribe dynamical rotational transport.

11 Covering Spaces and Monodromy

- Versions label rotational orbits in G/H . Charts give local coordinates.
- For methylamine, choose an H -invariant cell $\mathcal{U}$ whose translates partition $\mathcal{F}$ up to their walls. Glue orientations, normal coordinates, and wavefunctions across the walls.

$$(G_{12} \times_H \mathcal{U}) / \sim \cong \mathcal{F}.$$

- For a free action, $\mathcal{F} \rightarrow \mathcal{F}/G$ is a covering with deck group G . A lifted chart transition induced by g is a local expression of its deck transformation.

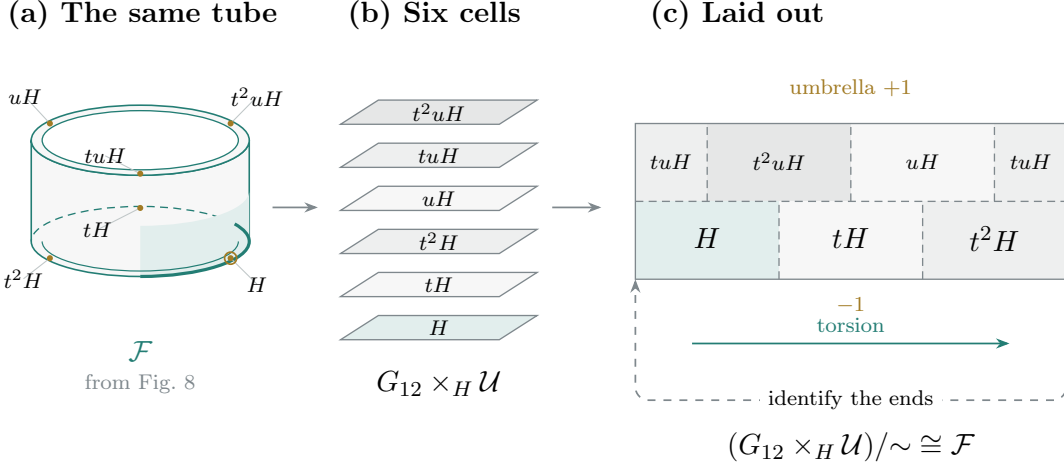


Figure 10: The six versions of Fig. 8, carried from the tube into cells. Teal marks the reference cell $\mathcal{U}$. The laid-out tube retains the staggered version positions, with the tuH cell split across the end seam. Dashed walls carry coordinate transitions, including the suppressed orientation and normal coordinates. Fig. 11 resolves two local covering sheets within each cell.

- A path $X \rightarrow gX$ projects to a loop. Its lift stays in one feasible component and records the sheet change, or *monodromy*.⁷
- Near a free reference configuration, twelve local sheets group in pairs within six version cells. G_{12}/H is a coset set, not a group.

$$t^3 = b^2 = E, \quad btb = t^{-1}, \quad bt = t^2b \neq tb.$$

- A species sees g through $\Gamma(g)$, with $\Gamma(G) \cong G/\ker \Gamma$. On G_6 , A_1 sees no change, while E retains noncommutativity.
- Treat fixed configurations separately. Recover the rigid orientation construction.⁸
- Distinguish chart gluing from mechanical parallel transport.⁹ A Berry interpretation needs a specified state family and connection.

⁷Hatcher [8], Section 1.3.

⁸Albert et al. [9].

⁹Littlejohn and Reinsch [10].

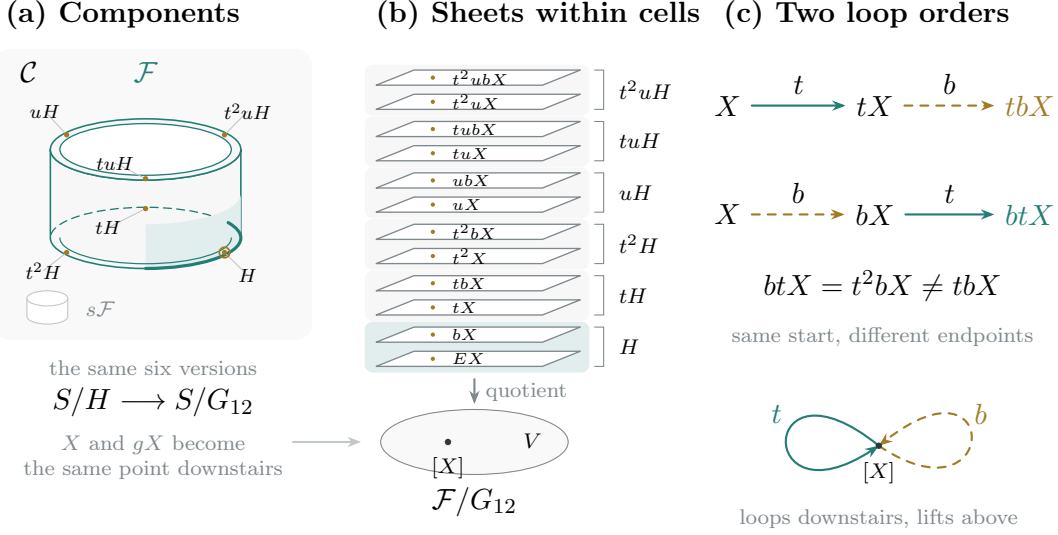


Figure 11: The G_{12} tube and version cells of Fig. 10, now viewed through a free covering. Each cell contains two local sheets over a small quotient neighborhood V . Orientations and normal coordinates are suppressed in the tube only. Loops t, b lift from X to tX, bX . Continuing from sX ends at stX, sbX , so order matters. Loop continuation acts on the right of the sheet label, while deck transformations act on the left.

12 Momentum Representation

- Expand the same states in rotational, internal, and normal modes, subject to the chart identifications.
- For the product model $SO(3) \times S^1 \times \mathbb{R}^d$, absorb the density into $\tilde{\psi} = \mathcal{J}^{1/2}\psi$. Use the measure $dr d\tau dQ$, denoted by the subscript 0.

$$\langle r, \tau, Q | JMK, m, p \rangle_0 = \frac{\sqrt{2J+1} D_{MK}^J(r) e^{im\tau} e^{ip \cdot Q/\hbar}}{\sqrt{2\pi} (2\pi\hbar)^{d/2}}.$$

$$J \in \mathbb{N}_0, \quad -J \leq M, K \leq J, \quad m \in \mathbb{Z}, \quad p \in \mathbb{R}^d.$$

- The coefficients $\hat{\psi}_{JMKm}(p)$ are spin vectors. Sum over J, M, K, m and integrate over p for the inverse transform. No energies are assigned. Check the product assumption.
- Let the body frame close with a turn while the normal frame closes on itself.

$$(r, \tau + 2\pi, Q) \sim (rR_z(-2\pi\rho), \tau, Q), \quad D_{MK}^J(rR_z(\alpha)) = e^{-iK\alpha} D_{MK}^J(r).$$

$$e^{2\pi i\kappa} = e^{2\pi i\rho K}, \quad \kappa = m + \rho K, \quad m \in \mathbb{Z}.$$

- Periodic and twisted frames give integer and shifted labels for the same states. Include normal-frame gluing when present.
- For periodic scalar torsion, take $f_m(\tau) = e^{im\tau}$, with $t : \tau \mapsto \tau + 2\pi/3$ and $b : \tau \mapsto -\tau$.

$$U_g f = f \circ g^{-1}, \quad U_t f_m = e^{-2\pi im/3} f_m, \quad U_b f_m = f_{-m}.$$

- Find the species and spin partners of the modes. Extend to methylamine's coupled torsion and inversion.

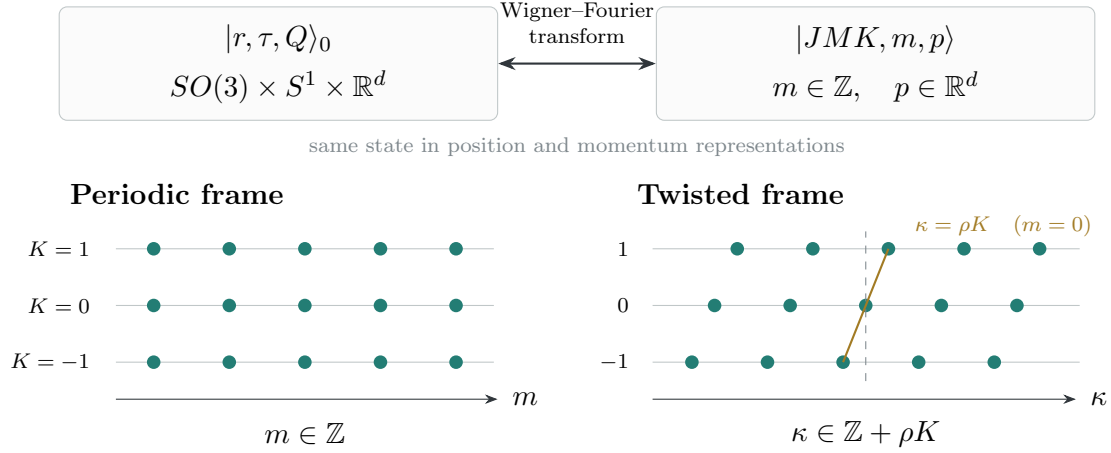


Figure 12: Position and momentum bases for the product model, with the chart density absorbed. Each row fixes K with $J \geq |K|$. The seam shifts the labels by ρK while preserving the state. The shear is schematic, and M, p are suppressed.

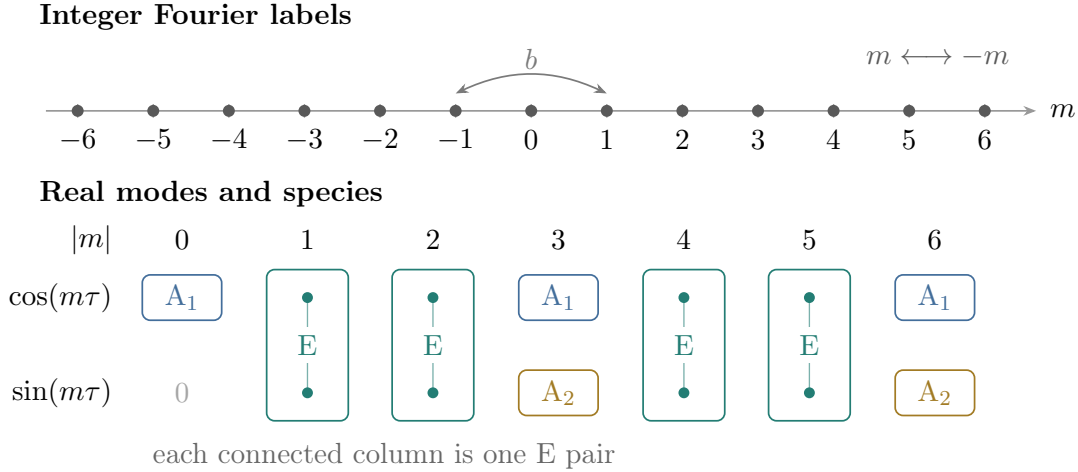


Figure 13: Periodic torsion modes grouped into species of G_6 . The constant belongs to A_1 . For each nonzero multiple of three, the cosine and sine belong to A_1 and A_2 . Every other pair spans E .

References

- [1] P. R. Bunker, C. J. H. Schutte, J. T. Hougen, I. M. Mills, J. K. G. Watson, and B. P. Winnewisser. Notations and conventions in molecular spectroscopy: Part 3. permutation and permutation-inversion symmetry notation (IUPAC recommendations 1997). *Pure and Applied Chemistry*, 69:1651–1657, 1997.
- [2] H. C. Longuet-Higgins. The symmetry groups of non-rigid molecules. *Molecular Physics*, 6:445–460, 1963.
- [3] Ming-Guang Hu, Yu Liu, Matthew A. Nichols, Lingbang Zhu, Goulven Quémener, Olivier Dulieu, and Kang-Kuen Ni. Nuclear spin conservation enables state-to-state control of ultracold molecular reactions. *Nature Chemistry*, 13:435–440, 2021.

- [4] P. R. Bunker and Per Jensen. *Molecular Symmetry and Spectroscopy*. NRC Research Press, Ottawa, second edition, 1998.
- [5] V. V. Ilyushin, E. A. Alekseev, S. F. Dyubko, R. A. Motiyenko, and J. T. Hougen. The rotational spectrum of the ground state of methylamine. *Journal of Molecular Spectroscopy*, 229(2):170–187, 2005.
- [6] I. Gulaczyk and M. Kręglewski. The effective Hamiltonian for the coupling between inversion–torsion states in methylamine. *Journal of Molecular Spectroscopy*, 256:86–90, 2009.
- [7] I. Kleiner and J. T. Hougen. A hybrid program for fitting rotationally resolved spectra of floppy molecules with one large-amplitude rotatory motion and one large-amplitude oscillatory motion. *Journal of Physical Chemistry A*, 119:10664–10676, 2015.
- [8] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002. Section 1.3.
- [9] Victor V. Albert, Eric Kubischta, Mikhail Lemeshko, and Lee R. Liu. Quantum theory of molecular orientation. Working manuscript.
- [10] Robert G. Littlejohn and Matthias Reinsch. Gauge fields in the separation of rotations and internal motions in the n-body problem. *Reviews of Modern Physics*, 69:213–275, 1997.